

WEB-BASED AIRPLANE RESERVATION SYSTEM

A PROJECT REPORT

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ABSTRACT

The primary objective of this project is to design and implement a comprehensive web-based airplane reservation system that automates and streamlines the process of flight booking and management. The system provides an intuitive platform for users to search for flights based on parameters such as source, destination, and date of travel, check seat availability, and perform ticket reservations in an efficient and reliable manner. The application is developed using modern web technologies, incorporating a responsive frontend and a backend integrated with a relational database to ensure efficient data storage, retrieval, and management. Additionally, the system includes important features such as booking cancellation, refund processing, and data validation mechanisms to enhance reliability and user satisfaction. Security measures such as controlled access and session management are implemented to protect user information. The modular architecture of the system ensures scalability and flexibility, allowing future enhancements such as integration with real-time airline APIs, payment gateways, and notification systems. The system consists of multiple functional modules including user registration and authentication, flight search and filtering, seat selection, ticket booking, and administrative controls. The admin module plays a crucial role in maintaining system integrity by allowing the management of flight schedules, updating seat capacities, and monitoring user activities.. Furthermore, the system includes features such as booking cancellation, data validation, and secure access control to maintain system reliability and user data protection. The design follows a modular and scalable architecture, making it adaptable for future enhancements such as and integration with external airline services. The airplane reservation system demonstrates how web-based solutions can improve the efficiency, accuracy, and convenience of airline booking processes, providing a practical and scalable approach to modern travel management. In conclusion, the airplane reservation system demonstrates how web-based applications can significantly improve the efficiency, accuracy, flexibility and usability of airline booking processes.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
API	Application Programming Interface
OTP	One Time Password
UI	User Interface
SRS	Software Requirement Specification
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DB	Database
SQL	Structured Query Language
HTTP	Hyper Text Transfer Protocol

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Air transportation plays a vital role in connecting people across cities, countries, and continents. With the increasing demand for air travel, efficient management of flight bookings has become essential. An airplane reservation system provides a digital solution to these challenges by allowing users to book tickets, check flight schedules, and manage reservations through an online platform. It simplifies the booking process, reduces paperwork, and ensures better accuracy in maintaining passenger and flight data. This project proposes a web-based Airplane Reservation System that enables users to search for flights based on source, destination, and travel date, check seat availability, and reserve tickets efficiently. The system is built using modern technologies, with a structured frontend interface and a backend connected to a relational database for storing user details, flight information, and booking records. The system includes key functionalities such as user registration and login, flight search, seat selection, ticket booking, and booking history management. An administrative module is also included, which allows the admin to manage flight schedules, update seat availability. The system updates seat availability after each booking, ensuring that users are provided with accurate information during the reservation process. By automating the booking system, this project reduces manual workload, minimizes errors, and enhances user convenience.



Figure 1.1 Airport Passenger Flow & Demand

1.1 OBJECTIVE

The primary objective of this project is to develop a functional, efficient, and user-friendly airplane reservation system that simplifies the process of flight booking and management. The specific objectives are:

1. To develop a web-based application that allows users to register, log in, and manage their accounts securely.
2. To implement a flight search system where users can find flights based on source, destination, and travel date.
3. To design a seat booking module that allows users to select available seats and confirm reservations.
4. To provide a booking management system where users can view, cancel, or modify their reservations.
5. To develop an admin module to manage flight schedules, update seat availability, and monitor system activities.
6. To create a simple and intuitive user interface for smooth interaction and better user experience.

1.3 PROBLEM DESCRIPTION

The key problems addressed in this project are as follows:

1. Traditional flight booking systems often involve manual processes that are time-consuming and inefficient.
2. Lack of a centralized system leads to difficulties in managing flight schedules and passenger records.
3. Higher chances of human errors such as incorrect bookings and duplicate seat allocation.
4. Difficulty for users in accessing real-time or updated information about flight availability.
5. Limited accessibility and inconvenience in booking tickets without an online platform.
6. Inefficient handling of booking cancellations and record maintenance.
7. Lack of a structured system to manage large volumes of passenger and flight data effectively.

1.4 MOTIVATION

With the rapid growth of air travel, managing flight bookings efficiently has become increasingly important. Traditional booking methods often involve long queues, manual errors, and lack of instant access to flight information, making the process inconvenient for passengers. There is a strong need for a reliable and automated system that simplifies ticket booking and ensures accuracy.

The motivation behind this project is to develop a practical and scalable airplane reservation system that provides users with an easy and efficient way to book flights. By automating the reservation process and maintaining a centralized database, the system reduces human errors, improves overall efficiency. It enhances user experience by providing access to flight details and booking options.

The key motivating factors for this project include:

- Increasing demand for online and digital booking systems.
- Need to reduce manual errors and improve data accuracy.
- Requirement for a centralized system to manage flight and passenger records.
- Growing use of web applications for travel and booking services.
- Opportunity to apply programming and database management skills to a real-world problem.

1.5 ORGANIZATION OF THE REPORT

This report is organized into multiple chapters. Chapter 1 presents the introduction, including overview, objectives, problem description, and motivation. Chapter 2 discusses the literature review and related work in airline reservation systems. Chapter 3 covers system analysis, including existing system, proposed system, and software requirements. Chapter 4 explains the system design, including architecture diagrams, data flow diagrams, and database design. Chapter 5 describes the system implementation, including coding details and module development. Chapter 6 presents system testing, including unit testing, integration testing, and validation. Chapter 7 demonstrates the system outputs with relevant screenshots. Chapter 8 concludes the project and discusses future enhancements. References and appendices are included at the end of the report.

CHAPTER 2

RELATED WORK

2.1 LITERATURE REVIEW

O'Brien and Marakas [1] discussed that airline reservation systems are a type of management information system that integrates databases, transaction processing, and user interfaces to handle booking operations efficiently. Their work emphasized the importance of centralized data management and real-time transaction processing in such systems.

Patil et al. [2] developed a web-based airline reservation system using Java and MySQL, focusing on modules such as user authentication, flight search, ticket booking, and administrative control. Their study highlighted that automation reduces manual errors and improves system efficiency.

Sharma et al. [3] analyzed the role of online reservation systems in improving customer experience. They concluded that features such as online booking, seat availability checking, and cancellation significantly enhance user convenience and reduce processing time.

Kumar and Singh [4] studied database-driven reservation systems and emphasized the importance of maintaining data consistency and avoiding duplicate bookings through proper validation and transaction management techniques.

Barua et al. [5] proposed a scalable airline reservation system using microservices architecture. Their research demonstrated improved performance, flexibility, and the ability to handle multiple users simultaneously without system failure.

Zheng et al. [6] explored the use of blockchain technology in airline reservation systems to enhance security and transparency. Their study showed that decentralized systems can prevent data tampering and improve trust in booking transactions.

Overall, the literature indicates that modern airline reservation systems focus on automation, centralized data management, scalability, and security. The proposed system aims to address these gaps by developing an efficient and accessible web-based airplane reservation system.

2.2 EXISTING TECHNOLOGY

Existing airline reservation and ticket booking services are widely available through popular online travel platforms such as MakeMyTrip, Goibibo, and Yatra. These platforms provide users with the ability to search for flights, compare ticket prices, check seat availability, and book tickets through web and mobile applications.

These systems are designed for large-scale public use and offer advanced features such as real-time booking, fare comparison, multiple airline options, online payment integration, and booking management. They provide a smooth and user-friendly interface, making it convenient for users to plan and manage their travel efficiently. In addition to these platforms, individual airline websites such as IndiGo and Air India also provide direct booking services. These systems allow users to book tickets, select seats, check flight status, and manage their reservations.

However, these existing systems are highly complex and designed for commercial-scale operations. They require strong backend infrastructure, secure payment gateways, and integration with multiple airline databases. Due to their complexity, they are not suitable for small-scale or academic implementations.

Therefore, there is a need for a simpler and more focused system that captures the core functionalities of flight booking while remaining easy to design and implement.



Figure 2.1 Existing Technology

2.3 INFERENCE OF LITERATURE REVIEW

The literature review on airline reservation systems highlights key aspects required for efficient booking and management. Existing systems focus on seat allocation, passenger record management, and scheduling using database-driven approaches.

Studies show that real-time booking systems improve user experience by providing immediate seat availability and confirmation. However, some systems lack proper administrative controls and role-based access, which are essential for managing flights, users, and bookings securely.

Research also emphasizes the importance of transparency in ticket pricing, cancellation policies, and booking status updates. Identity verification and secure transaction mechanisms play a crucial role in enhancing passenger trust and reducing fraudulent bookings.

Additionally, incorporating features like automated seat allocation, booking history tracking, and notification systems improves overall efficiency and usability. Overall, the literature suggests that an effective airline reservation system should integrate real-time booking, security features, and user-friendly interfaces to ensure reliability and efficiency.

Table 2.1 Inference of Literature Review

Title	Contribution
Optimization of Airline Scheduling Systems	Proposed models for efficient flight scheduling and seat allocation.
Survey of Airline Reservation Systems	Identified key features like booking management, cancellation, and pricing strategies.
Real-Time Airline Booking Systems	Highlighted importance of instant seat confirmation and availability tracking.
Airport Management Systems	Showed need for administrative control and role-based access..
Online Ticket Booking Systems Review	Emphasized transparency in pricing and booking processes.
Passenger Management Systems	Suggested identity verification and secure booking to reduce fraud.

CHAPTER 3

SYSTEM ANALYSIS

3.1 PROBLEM DEFINITION

Passengers often face difficulties in booking flight tickets efficiently due to limitations in traditional and existing booking systems. Manual booking methods are time-consuming, prone to human errors, and lack proper data management. Even with the availability of online booking platforms, users may encounter issues such as complex interfaces, lack of centralized control, and difficulty in managing bookings.

Existing systems are primarily designed for large-scale commercial use and may not be suitable for simplified or academic-level implementations. There is a need for a system that provides essential booking functionalities in a structured, user-friendly, and efficient manner.

Key limitations in existing solutions include:

1. Lack of a simple and centralized platform for managing flight bookings.
2. Possibility of errors such as duplicate seat booking and incorrect data entry.
3. Difficulty in maintaining and accessing booking records efficiently.
4. Absence of a streamlined system for managing flight schedules and seat availability.
5. Limited user control over booking management such as cancellation and history tracking.

3.2 PROPOSED SOLUTION

The proposed Airplane Reservation System is a web-based application that provides a structured and efficient platform for managing flight bookings. The system incorporates advanced features such as user-type classification, risk-based pricing, gamification, and weather-based insights to enhance user experience and decision-making.

The system includes the following key components:

1. User Module:

Allows users to register, log in, and book flights based on their preferences.

2. Flight Search Module:

Enables users to search for flights using parameters such as source, destination, and travel date.

3. Seat Booking Module:

Allows users to view available seats and reserve them. Seat availability is updated after each booking to maintain consistency.

4. User Type Classification:

The system categorizes users based on travel purpose such as:

- a. Business Class Users (priority, less risk tolerance)
- b. Vacation Travelers (flexible, cost-sensitive)
- c. Economy Travelers (budget-focused)

This classification helps in customizing recommendations and pricing strategies.

5. Risk Management System:

The system evaluates flight-related risks based on factors such as weather conditions, historical delays, and travel congestion. Each flight is labeled as:

- a. Safe
- b. Moderate Risk
- c. High Risk

This helps users make informed decisions before booking.

6. Dynamic Pricing with Risk-Based Discounts:

The system offers lower ticket prices for higher-risk flights, encouraging users to choose flexible options. Users willing to take higher risks (such as possible delays) can benefit from discounted fares, making the system cost-efficient.

7. Gamification Feature:

The system enhances user interaction by providing feedback such as:

- a. “Smart Choice”
- b. “Budget Saver”

c. “High Risk – Low Cost”

This makes the booking experience more engaging and informative.

8. Weather Forecast Integration:

The system provides weather insights for the selected travel date and destination, helping users anticipate possible delays or disruptions.

9. Booking Management Module:

Allows users to view booking details, cancel reservations, and track booking history.

10. Admin Module:

Enables the administrator to manage flights, update pricing, monitor risks, and control system operations.

11. Database Management:

Stores all user, flight, booking, pricing, and risk-related data efficiently.

12. Validation and Security:

Ensures secure login, proper data validation, and prevention of unauthorized access.

3.3 SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.3.1 Functional Requirements

- User Registration: Users can register with details such as name, email, password.
- User Login: Registered users can log in securely to access the system.
- Admin Login: Admin can log in using credentials to access the admin panel.
- Flight Management: Admin can add, update, and delete flight details including source, destination, schedule, and seat capacity.
- Flight Search: Users can search for available flights based on source, destination, and travel date.
- Seat Selection: Users can view available seats and select preferred seats during booking.
- Ticket Booking: Users can book tickets and receive booking confirmation.
- Booking Management: Users can view booking history, cancel tickets, and manage reservations.
- User Type Classification: The system categorizes users as Business, Vacation, or Economy travelers to customize recommendations and pricing.
- Risk Management: The system analyzes flights and assigns risk levels such as Safe, Moderate, or High Risk based on factors like weather and historical delays.

- **Dynamic Pricing:** Ticket prices are adjusted based on risk level, where higher-risk flights offer lower fares.
- **Gamification Feature:** The system provides feedback such as “Best Deal”, “Safe Choice”, or “High Risk – Low Cost” to guide users.
- **Weather Information:** The system displays weather conditions for the selected travel date and destination.
- **Data Storage:** All user, flight, and booking details are stored in a database.

3.3.2 Non-Functional Requirements

- **Security:** The system ensures secure login and protects user data through authentication mechanisms.
- **Performance:** The system should respond quickly to user requests under normal load conditions.
- **Usability:** The interface should be simple, user-friendly, and easy to navigate.
- **Reliability:** The system should maintain accurate data and operate without failure.
- **Scalability:** The system should handle an increasing number of users and bookings efficiently.
- **Maintainability:** The system should be easy to update and maintain.

3.4 SOFTWARE REQUIREMENTS

Programming Language – Java

Java is used as the primary programming language for developing the airplane reservation system. It provides object-oriented features, platform independence, and robust performance for building reliable applications.

Development Environment – Visual Studio Code / IntelliJ IDEA

The project is developed using an Integrated Development Environment (IDE) such as Visual Studio Code or IntelliJ IDEA, which provides tools for coding, debugging, and project management.

User Interface – Java (Console / Swing)

The system interface is developed using Java, either as a console-based application or using Java Swing for graphical user interface components.

Authentication and Security

Basic authentication is implemented using username and password validation. Passwords can be securely stored using hashing techniques to improve security.

Operating System

The system can run on any operating system such as Windows, Linux, or macOS that supports Java.

3.5 USE CASES

Admin Use Cases

- Login to the admin panel using username and password.
- Add, update, or delete flight details including source, destination, schedule, and seat capacity.
- View and manage all registered users.
- Monitor all bookings and flight activities.
- Update ticket pricing and manage risk-based pricing configurations.
- Set or adjust risk levels (Safe, Moderate, High Risk) for flights.
- Manage system data including flights, bookings, and user classifications.

User Use Cases

- Register with personal details and log in to the system.
Search for flights by entering source, destination, and travel date.
- View flight details including available seats, price, and risk level.
- Select seats and proceed with ticket booking.
- View weather information for the selected travel date and destination.
- Receive system suggestions such as:
 - “Safe Choice”
 - “Budget Option”
 - “High Risk – Low Cost”

- Book tickets based on preference (safe vs cheaper risky flights).
- View booking confirmation and booking history.
- Cancel bookings if required.

System Use Cases (Internal Functions)

- Classify users into Business, Vacation, and Economy travelers.
- Analyze flight conditions to assign risk levels based on predefined factors.
- Apply dynamic pricing, offering lower fares for higher-risk flights.
- Update seat availability after each booking.
- Store and retrieve all data from the database efficiently.
- Validate user inputs and ensure secure access to the system.

CHAPTER 4

SYSTEM DESIGN

4.1 INTRODUCTION

The system design chapter describes the overall architecture, module breakdown, and data flow of the proposed Airline Reservation System. The system follows a modular architecture implemented using Java, ensuring scalability, reliability, and ease of maintenance. The design focuses on providing an efficient booking experience, integrating features such as seat layout visualization, weather-based pricing, and secure payment processing.

4.2 OVERALL SYSTEM ARCHITECTURE

The system is organized into three major layers:

1. **Presentation Layer:**

A user-friendly interface (console/GUI-based Java application) that allows users to search flights, view seat layouts, select travel classes (Economy, Business, Vacation), and perform bookings. It also displays notifications and booking confirmations.

2. **Application Layer:**

Core logic implemented in Java handles all functionalities such as flight search, seat allocation, weather-based risk analysis, dynamic pricing, ticket generation, and payment processing. It ensures validation of user inputs and manages booking workflows.

3. **Data Layer:**

The system uses file handling or a database to store data such as user details, flight schedules, seat availability, bookings, and pricing information. It ensures efficient data retrieval and updates during booking operations.

The workflow begins when a user interacts with the system to search for available flights by entering source and destination details. The system retrieves matching flights along with seat availability and pricing.

For the booking flow, the user selects a flight → chooses a seat from the seat layout → selects travel class → the system evaluates weather conditions to determine risk level → pricing is

dynamically adjusted (lower price for high-risk conditions) → the user proceeds with payment → upon successful payment, a ticket is generated and stored → a confirmation notification is sent to the user.

For the weather-based pricing flow, the system analyzes weather conditions of the departure and arrival locations → classifies them into safe or risk categories → if the condition is risky, ticket prices are reduced to attract bookings → otherwise, normal pricing is applied. For the admin management flow, the administrator can update flight schedules, modify pricing rules, and monitor bookings → changes are reflected immediately in the system → ensuring updated and accurate information for users.

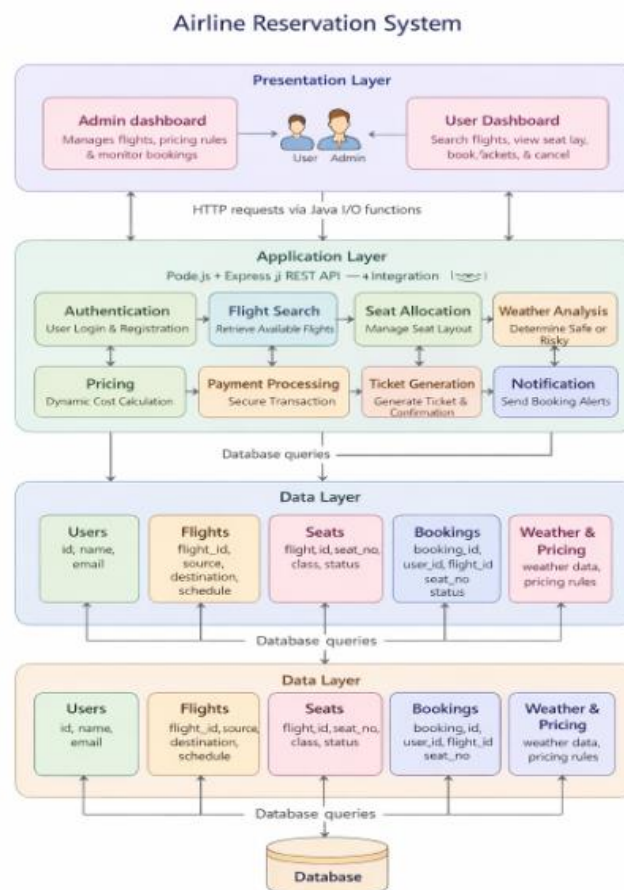


Figure 4.1 System Architecture

4.3 SYSTEM WORKFLOW

The end-to-end system workflow follows these steps:

1. A new user registers and logs into the system. The system stores user details such as name, email, and login credentials.
2. The user searches for flights by entering source, destination, and travel date. The system retrieves matching flights along with available seats and pricing details.
3. The user selects a flight and views the seat layout. Available seats are displayed, and the user selects a preferred seat along with travel class (Economy, Business, or Vacation).
4. The system analyzes weather conditions for the selected route and classifies them as safe or risky. If the condition is risky, the ticket price is reduced dynamically; otherwise, normal pricing is applied.
5. The user proceeds to booking by confirming flight details and entering passenger information.
6. The system redirects to the payment module, where the user completes the transaction using available payment methods.
7. Upon successful payment, the system generates a ticket with booking details such as flight number, seat number, class, and timing. The ticket is stored in the system.
8. A confirmation notification is sent to the user regarding successful booking. The user can also view booking history in their dashboard.
9. The user has the option to cancel the booking. If canceled, the system updates the booking status and records the cancellation reason.
10. In case of system or payment failure, appropriate error messages are displayed and the user is prompted to retry.
11. The system ensures data security by validating user inputs and protecting transaction details.
12. The user can view, download, or print the ticket after successful booking.

generation. The module ensures secure access using password hashing and session management techniques.

2. Admin Login

The admin login process is straightforward. Admin credentials are pre-stored in the system database with encrypted passwords. When the admin enters login details, the system retrieves the corresponding record and verifies the password. If the credentials are valid, access is granted to the admin dashboard, where the admin can manage flights, update pricing rules, and monitor bookings.

3. User Registration and Login

User registration allows new passengers to create an account by providing details such as name, email, phone number, and password. The password is securely stored using hashing techniques to ensure data protection. During login, the system verifies user credentials by comparing the entered password with the stored encrypted password. If authentication is successful, the user is granted access to search flights, view seat layouts, and make bookings.

4. Session Management

Once logged in, the system maintains a user session to allow continuous interaction without repeated authentication. This enables smooth navigation across modules such as flight search, booking, and payment.

5. Security Mechanisms

The system ensures security through encrypted password storage and validation checks. User inputs are validated to prevent unauthorized access. Only authenticated users can perform booking operations, while admin privileges are restricted to authorized personnel.

6. Admin Module:

The Admin Module acts as the central control unit of the Airline Reservation System. All administrative functionalities are restricted to authorized personnel, ensuring that only admins can manage flights, pricing, and system operations. The admin does not participate in booking directly but controls and monitors all activities within the system.

Flight Management

One of the primary responsibilities of the admin is managing flight details. The admin can add new flights by entering information such as flight number, source, destination, departure time, arrival time, and total seats. These details are stored in the system and made available for users during flight search. The admin can also update or delete existing flights. Any modification made is immediately reflected in the system, ensuring that users always see up-to-date flight information. This helps maintain accuracy and prevents booking conflicts.

Seat Configuration

The admin is responsible for defining seat layouts for each flight. Seats are categorized into different classes such as Economy, Business, and Vacation. The system tracks seat availability and updates it dynamically as bookings are made.

This module ensures that seat allocation is properly managed and prevents overbooking by restricting users from selecting already booked seats.

Pricing Management

The admin controls ticket pricing across the system. Base fare values for different travel classes are defined and stored in the system. The admin can update these values at any time depending on demand, season, or operational requirements.

In addition to base pricing, the system includes a weather-based pricing mechanism. The admin can configure rules that adjust ticket prices when weather conditions are classified as risky. For example, during bad weather, ticket prices may be reduced to encourage bookings. All pricing changes are applied instantly, and users see updated fares during flight search and booking.

Booking Monitoring

The admin has full visibility over all bookings made in the system. This includes passenger details, flight information, seat number, booking status, and payment status. The admin can track whether bookings are active, completed, or canceled. This helps in monitoring system usage and resolving any issues related to bookings.

User Management

The admin can view all registered users in the system. While sensitive data such as passwords are protected, basic details like name and contact information are accessible for monitoring purposes. This module helps identify suspicious activity and ensures that only valid users are interacting with the system.

Notification and Alerts

The admin module also supports system-wide notifications. Admins can trigger alerts related to flight updates, delays, cancellations, or important announcements. These notifications are sent to users and displayed in their dashboard.

Payment and Refund Management

The Admin Module handles monitoring of payment transactions and refund processes. Admins can verify successful payments, track failed transactions, and initiate refunds for cancelled tickets. This ensures transparency and builds trust with users while maintaining accurate financial records.

Overall Role of the Admin Module

The Admin Module ensures proper functioning, control, and reliability of the Airline Reservation System. It maintains data accuracy, enforces pricing rules, manages flights and seats, and monitors all bookings. By centralizing control, the system ensures a secure and organized booking environment, preventing misuse and maintaining consistency across all operations.

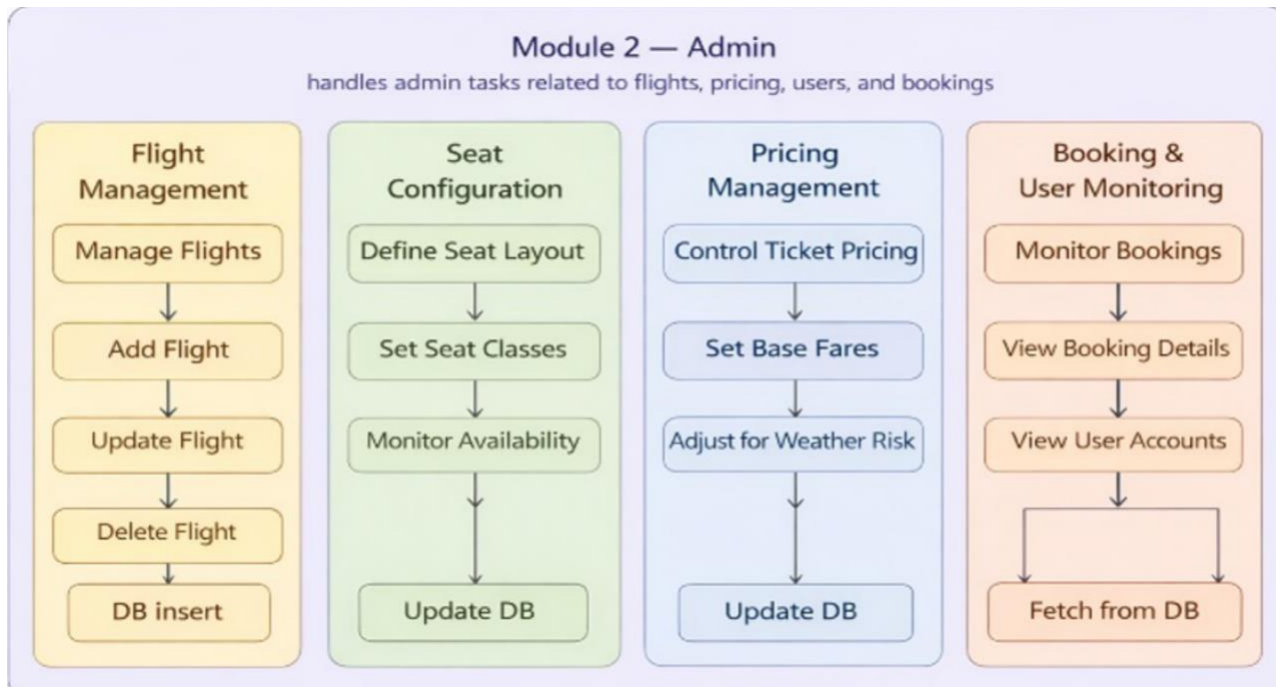


Figure 4.4 Admin Module

7. Customer Module:

The Customer Module is the core interaction layer of the Airline Reservation System. It is responsible for handling all user-side operations such as searching flights, booking tickets, making payments, and managing cancellations. Only registered and authenticated users can access this module, ensuring secure and personalized booking experiences.

Flight Search

Once a user logs into the system, the first action is searching for available flights. The user enters details such as source, destination, and travel date. The system processes this request and retrieves matching flights from the database along with seat availability, travel class options, and pricing details. The results displayed include flight number, departure and arrival time, available seats, and ticket cost. This allows the user to compare different options before proceeding with booking.

Seat Selection and Booking

After selecting a flight, the user is presented with a seat layout showing available and booked seats. The user selects a preferred seat and chooses a travel class such as Economy, Business, or Vacation. The system ensures that the selected seat is available in real time. Once confirmed, the booking process continues with the selected flight, seat, and class details.

Weather-Based Pricing

Before finalizing the booking, the system evaluates weather conditions for the selected route. Based on the analysis, the system classifies conditions as safe or risky. If the conditions are risky, the system reduces the ticket price dynamically to encourage bookings. If conditions are safe, standard pricing is applied. The final fare is displayed to the user before proceeding to payment.

Payment Processing

The user proceeds to the payment stage by entering required payment details. The system processes the transaction securely. If the payment is successful, the booking is confirmed. In case of payment failure, an error message is displayed and the user is given the option to retry the payment without losing booking details.

Ticket Generation and Confirmation

After successful payment, the system generates a ticket containing all booking details such as flight number, passenger details, seat number, travel class, and timing. A unique booking ID is assigned, and the ticket is stored in the system. The user receives a confirmation notification and can view the ticket in their dashboard.

Booking Cancellation

The customer also has the option to cancel a booking if required. The user selects the booking and provides a cancellation reason. The system updates the booking status to cancelled and stores the reason for future reference.

Overall Role of the Customer Module

The Customer Module enables users to interact seamlessly with the Airline Reservation System. It manages the complete lifecycle of a booking, from searching flights to ticket confirmation and cancellation. By integrating features like real-time seat selection, dynamic pricing, and secure payments, the module ensures a smooth, efficient, and user-friendly booking experience.

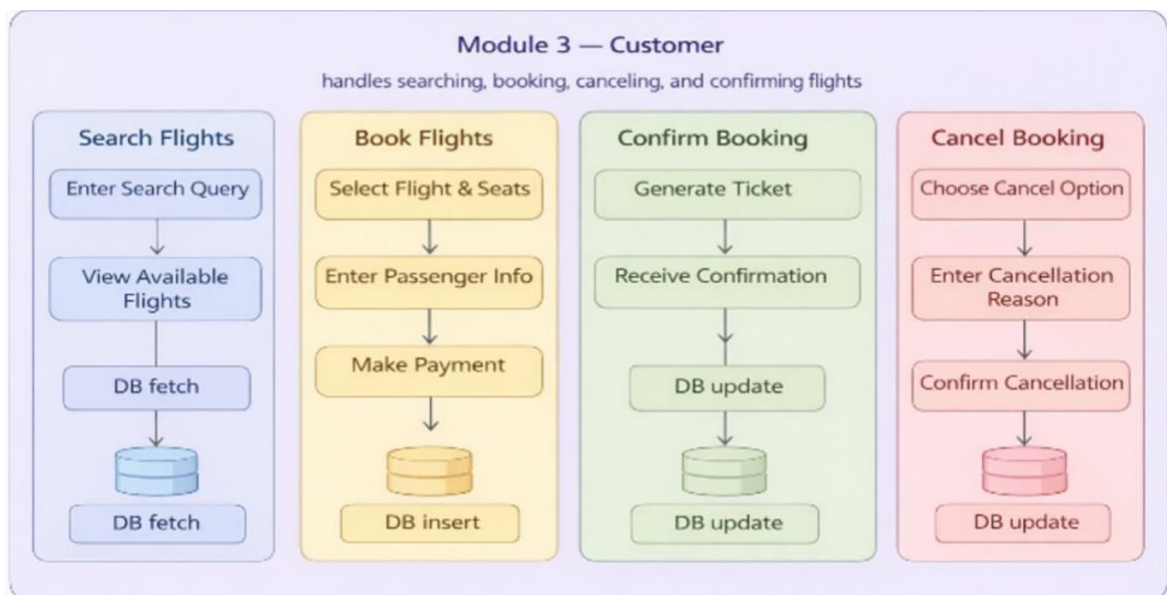


Figure 4.5 Customer Module

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 ALGORITHM USED IN THE PROJECT

The proposed Airline Reservation System uses a combination of user authentication, flight generation logic, seat allocation, dynamic pricing, and ticket generation. The implementation is designed using Java for backend logic and HTML, CSS, and JavaScript for frontend interaction.

The system is modular, where each component handles a specific task such as login, booking, pricing, and payment simulation.

5.2 ALGORITHM EXPLANATION

5.2.1 User Authentication Flow

- Step 1: User enters username and password in the login page.
- Step 2: The system checks localStorage for existing user data.
- Step 3: If the user exists, credentials are validated.
- Step 4: If valid, the username is stored as currentUser in localStorage.
- Step 5: User is redirected to the main booking page.
- Step 6: If new user, registration stores credentials in localStorage.

5.2.2 Flight Generation Algorithm

The system dynamically generates flights based on user input (source and destination).

- Step 1: User enters source, destination, and travel date.
- Step 2: The system creates multiple flight options with different timings.
- Step 3: Seat count is generated randomly.
- Step 4: Ticket price is generated within a range.
- Step 5: Flights are displayed to the user for selection.

5.2.3 Seat Allocation Algorithm

Step 1: User selects a flight.

Step 2: Seat layout (10 rows $\times$ 6 columns) is generated.

Step 3: Available seats are marked in green.

Step 4: Booked seats are disabled.

Step 5: User selects seats, which change to selected state.

Step 6: Selected seats are stored in an array.

5.2.4 Cost Calculation

5.2.4 Pricing Algorithm

Step 1: Base price = flight price $\times$ number of seats

Step 2: If booking type is Risk $\rightarrow$ price is reduced (30% discount)

Step 3: If purpose is Vacation $\rightarrow$ additional ₹500 added

Step 4: If purpose is Emergency $\rightarrow$ ₹1000 discount applied

Step 5: Final price is displayed before payment

5.2.5 Risk-Based Booking Logic

Step 1: User selects booking type (Safe or Risk)

Step 2: If Risk is selected:

- System applies discount
- Random outcome is generated

Step 3: Outcome can be:

- Upgrade (better seat)
- Last seat assignment

Step 4: Booking status is updated accordingly

5.2.6 Payment and Ticket Generation

Step 1: User enters card details (card number, name, CVV).

Step 2: System validates input fields.

Step 3: Payment is simulated with a delay.

Step 4: On success, booking is confirmed.

Step 5: Ticket is generated with details:

- Passenger name
- Route
- Date
- Seat numbers
- Price

Step 6: Ticket is displayed and can be downloaded.

5.2.7 Cancellation and Refund Logic

Step 1: User selects cancel ticket option.

Step 2: System calculates refund amount:

- Risk booking → 50% refund
- Safe booking → 80% refund

Step 3: Seats are restored to flight availability.

Step 4: User cancellation count is updated.

Step 5: Refund message is displayed.

5.3 EXPERIMENTAL SETUP

5.3.1 Software Configuration

- Programming Language: Java, JavaScript
 - Frontend: HTML, CSS
 - Storage: Local Storage (Browser), File Handling (Java)
 - IDE: VS Code
 - Browser: Brave

5.4 CODING IMPLEMENTATION DETAILS

5.4.1 Backend Structure (Java)

The backend is implemented using Java classes:

- AirlineReservation.java: main program logic
- Flight class: stores flight details (id, route, seats, price)
- Passenger class: stores user data (name, trips, cancellations, score)

File handling is used to store and retrieve user information.

5.4.2 Frontend Structure

The frontend consists of multiple HTML pages:

- home.html → landing page
- login.html → user authentication
- index.html → main booking interface
- support.html → user support page

JavaScript is used to handle:

- Flight generation
- Seat selection
- Pricing logic
- Payment simulation
- Ticket generation

CHAPTER 6

SYSTEM TESTING

6.1 GENERAL

System testing is performed to verify that all modules of the Airline Reservation System function correctly, both individually and as an integrated system. Testing ensures that user authentication, flight search, seat selection, booking, payment simulation, and ticket generation work as expected. The testing phases include unit testing, integration testing, functional testing, performance testing, and validation testing.

6.2 UNIT TESTING

The unit testing process was conducted to verify the core functionalities of the system, including user registration, login, flight viewing, and seat layout generation. Each test case was executed using appropriate input data and compared against the expected outcomes.

Table 6.1 Unit Testing

St	Test Case (Steps)	Test Data	Expected Result	Actual Result	Status
1	Register a new user	Username, email, password	User registered successfully	Registration successful	Pass
2	User Login	Valid username & password	Login successful	User redirected to booking page	Pass
3	View Flights	Source, destination	Flights displayed	Flights shown correctly	Pass
4	Generate seat layout	Flight selected	Seat Map displayed Successfully	Seat grid displayed	Pass

6.3 INTEGRATION TESTING

Integration testing was carried out to ensure that different modules of the system work together seamlessly. The interactions between login, flight search, seat selection, booking, payment, and cancellation modules were tested using valid input data. The results indicate that after successful login, users were able to search and view flights correctly based on selected routes.

The seat selection module properly highlighted and confirmed chosen seats for the selected flight. Furthermore, the booking and payment process generated tickets accurately without any issues. The cancellation and refund functionality also worked as expected, with refunds being processed and displayed correctly.

Since all actual results matched the expected outcomes, all integration test cases passed successfully, confirming that the system components are well-integrated and function smoothly as a whole.

Table 6.2 Integration Testing

St	Test Case (Steps)	Test Data	Expected Result	Actual Result	Status
1	User logs in and searches flights	Valid login + route	Flights displayed	Flights shown correctly	Pass
2	Select flight and choose seats	Flight + seat selection	Seats selected successfully	Seats highlighted correctly	Pass
3	Booking + payment flow	Valid details	Ticket generated	Ticket updated correctly	Pass
4	Cancel booking and refund	Bookinh exists	Refund processed	Refund shown correctly	Pass

6.4 FUNCTIONAL TESTING

Functional testing was performed to validate how the system handles incorrect inputs and edge cases. Various negative scenarios were tested, such as entering an incorrect password, leaving input fields empty, selecting the same source and destination, and attempting to book tickets without seat selection. The system responded appropriately in all cases by displaying relevant error messages or alerts. For instance, invalid login credentials triggered an error message, empty fields prompted a “fill all fields” alert, and selecting the same cities resulted in a warning to choose different locations.

Additionally, booking was successfully blocked when no seats were selected. Since all actual outcomes matched the expected results, all functional test cases passed, confirming that the system effectively handles invalid inputs and maintains proper validation throughout.

Table 6.3 Functional Testing

St	Test Case (Steps)	Test Data	Expected Result	Actual Result	Status
1	Login with wrong password	Incorrect password	Error message displayed	'Invalid credentials' shown	Pass
2	Empty input fields	No data entered	Login rejected with message	“Fill all fields” alert	Pass
3	Same source and destination	Same city selected	Error message	“Choose different cities” shown	Pass
4	Booking without seat selection	No seats selected	Booking blocked	Alert displayed	Pass

6.5 PERFORMANCE TESTING

Performance testing was conducted to evaluate the system's responsiveness and efficiency under normal operating conditions. Key operations such as user login, flight search, and payment processing were tested using valid input data. The results indicate that the system performs efficiently, with login responses completed in less than 500 milliseconds, which is well within the expected limit of 2 seconds.

Similarly, flight generation results were displayed in under 1 second, ensuring a smooth user experience during search operations. The payment simulation process was also completed in under 2 seconds, meeting the expected performance criteria.

Since all operations were executed faster than the expected response times, all performance test cases passed successfully, demonstrating that the system is optimized for speed and can handle user interactions effectively.

Table 6.4 Performance Testing

St	Test Case (Steps)	Test Data	Expected Result	Actual Result	Status
1	Login response time	Valid credentials	Response within 2 seconds	Response in < 500ms	Pass
2	Flight generation time	Route input	Results within 2 seconds	Results in < 1 second	Pass
3	Payment simulation	Valid inputs	Payment processed instantly	generated in < 2 seconds	Pass

6.6 VALIDATION TESTING

Validation testing was performed to ensure that the complete system meets user requirements and functions correctly in real-world scenarios. End-to-end workflows such as booking a ticket—from login and flight search to payment—were tested, and the system successfully generated tickets without any issues. The price calculation feature was also validated by using different inputs, and the system accurately updated and displayed the revised costs to users.

Additionally, the cancellation and refund process worked correctly, with refunds being processed and shown as expected. Since all actual results matched the expected outcomes, all validation test cases passed successfully, confirming that the system is reliable, user-friendly, and ready for deployment.

Table 6.5 Validation Testing

St	Test Case (Steps)	Test Data	Expected Result	Actual Result	Status
1	End-to-end booking flow	Login, search, book, pay	Ticket generated	Full flow completed	Pass
2	Price calculation check	Different purpose and risk	New cost shown to users	Price updated correctly	Pass
3	Cancellation and refund	Cancel booking	Refund displayed	Refund correct	Pass

CHAPTER 7

OUTPUT AND EXPLANATION

7.1. HOME PAGE OUTPUT

The above figure shows the home page of the Airline Reservation System. It acts as the main entry point for users accessing the application. The page contains a navigation bar with options such as Home, Support, and Login, allowing users to move between different sections easily. At the center of the page, a welcoming message “Fly Smart. Fly Easy.” is displayed along with a short description encouraging users to book flights and travel efficiently. A “Get Started” button is provided, which redirects the user to the login page to begin the booking process. The design of the home page is simple, visually appealing, and user-friendly, ensuring easy navigation and a smooth start to the system.

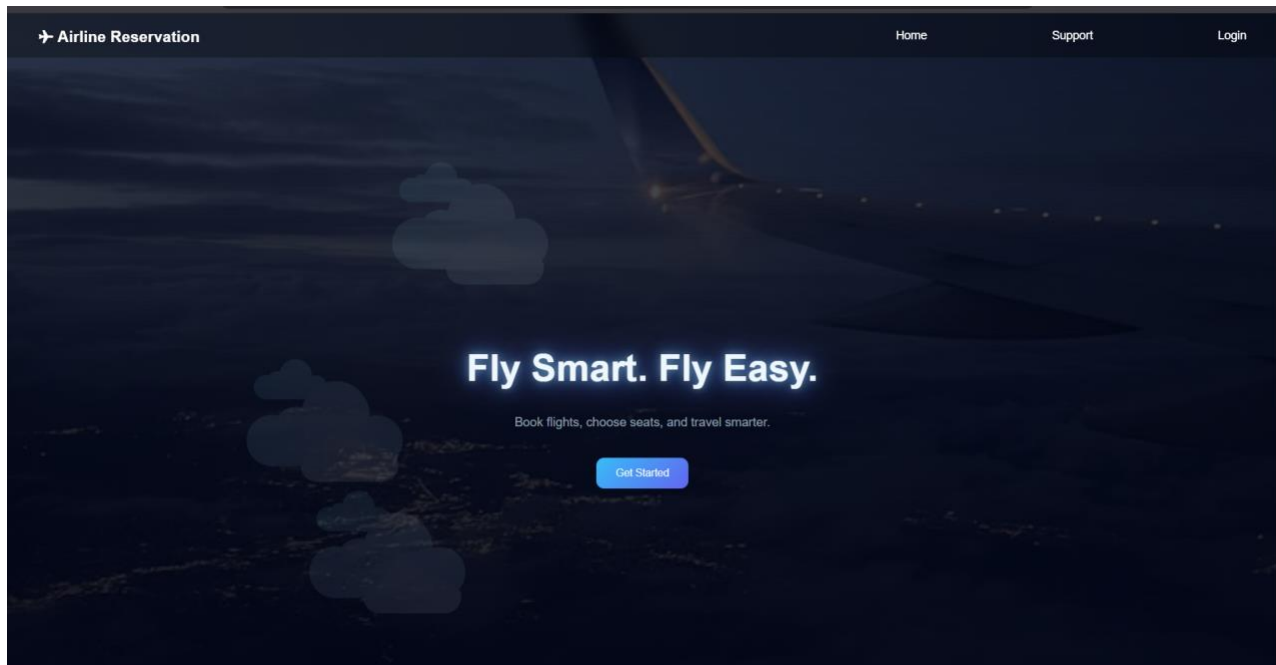


Figure 7.1 Home page

7.2 REGISTRATION / LOGIN PAGE

7.2.1 Registration Page

The figure 7.2.1 shows the registration page of the Airline Reservation System, which is used by new users to create an account. If a user does not already have login credentials, they can switch from the login mode to the registration mode using the option provided on the screen. The page contains input fields where the user enters a username and password. When the registration process is completed, the system stores the user credentials in localStorage, allowing the user to log in later using the same details. The interface is designed to be simple and user-friendly, ensuring that new users can easily create an account without any complexity. This module plays an important role in enabling secure access to the system by maintaining user-specific data.

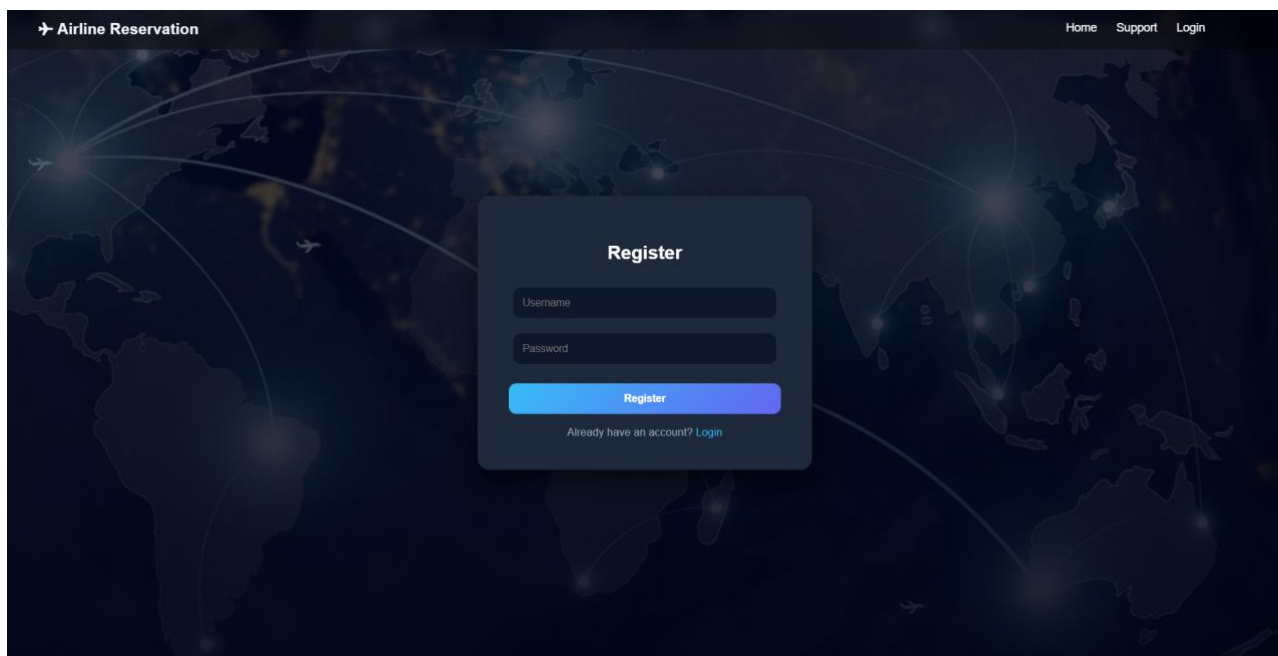


Figure 7.2.1 Registration page

7.2.2 Login Page

The figure 7.2.2 shows the login page of the Airline Reservation System, which is used by registered users to access the application. After completing the registration process, users can enter their username and password in the provided fields to log in.

Once valid credentials are entered, the system verifies the details using data stored in localStorage. If the authentication is successful, the user is redirected to the main booking interface where flight search and booking operations can be performed. The page also provides an option for new users to switch to the registration page if they do not have an account. The simple and secure login mechanism ensures that only authorized users can access the system functionalities.

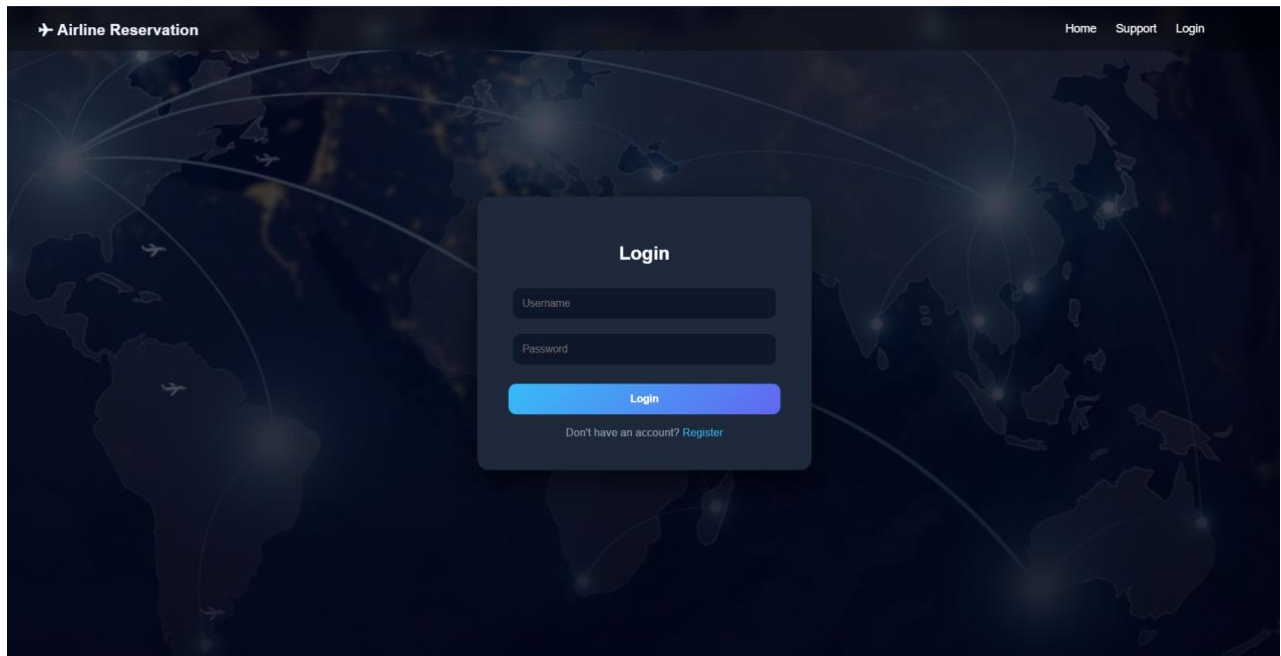


Figure 7.2.2 Login page

7.3 Flight Search Input Page

The figure 7.3 shows the flight search input page of the Airline Reservation System, which is displayed after successful login. This page collects essential travel details from the user, including the passenger name, source city, destination city, and travel date.

The interface uses dropdown menus for selecting cities, which helps in avoiding invalid inputs and ensures consistency in data entry. A date picker is provided to allow users to conveniently choose their preferred travel date, with restrictions applied to prevent selecting past dates. The design is simple and centered, making it easy for users to focus on entering the required information.

Once all the fields are filled correctly, the user clicks the “Start” button. The system then validates the entered data, ensuring that all fields are completed and that the source and destination are not the same. After validation, the system proceeds to generate available flights dynamically based on the selected route and date.

This page plays a crucial role in initiating the booking process, as it connects user input with the backend logic responsible for flight generation and further booking steps.

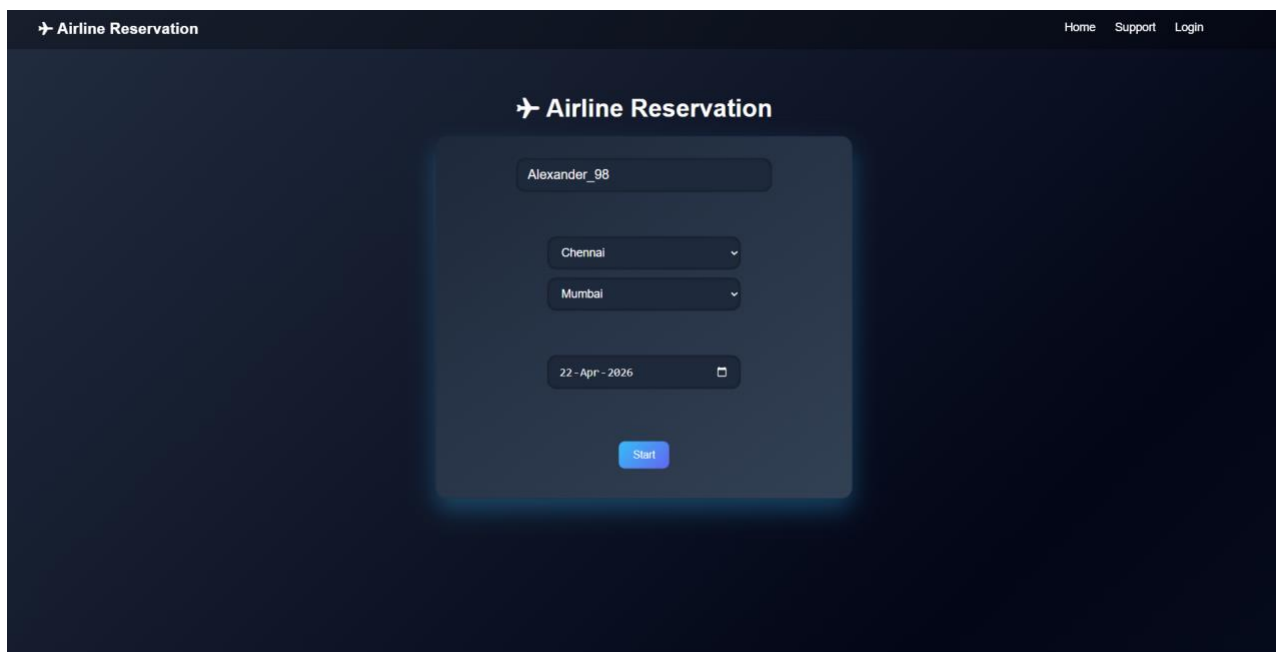
The image shows a web application interface for an airline reservation system. At the top, there is a dark blue header with the text "Airline Reservation" on the left and "Home Support Login" on the right. Below the header, the main content area has a dark blue background. In the center, there is a lighter blue rounded rectangle containing a search form. The form has a text input field with "Alexander_98", two dropdown menus for "Chennai" and "Mumbai", a date input field with "22-Apr-2026", and a blue "Start" button at the bottom.

Figure 7.3 Flight Search Page

7.4 Flight Selection Page

The figure 7.4 shows the flight selection page of the Airline Reservation System. After entering the travel details, the system displays available flights for the selected route, along with relevant information such as departure time, arrival time, number of available seats, and ticket price. At the top of the page, a welcome message is displayed along with the selected route, providing context to the user. The system also shows weather information for the destination city on the chosen date, adding realism to the booking process.

Multiple flight options are generated dynamically with different timings and pricing. Each flight is presented in a card format with a “Select” button, allowing the user to choose a preferred flight. Once a flight is selected, the system proceeds to the seat selection stage. This page plays an important role in helping users compare available options and choose the most suitable flight based on time and cost.

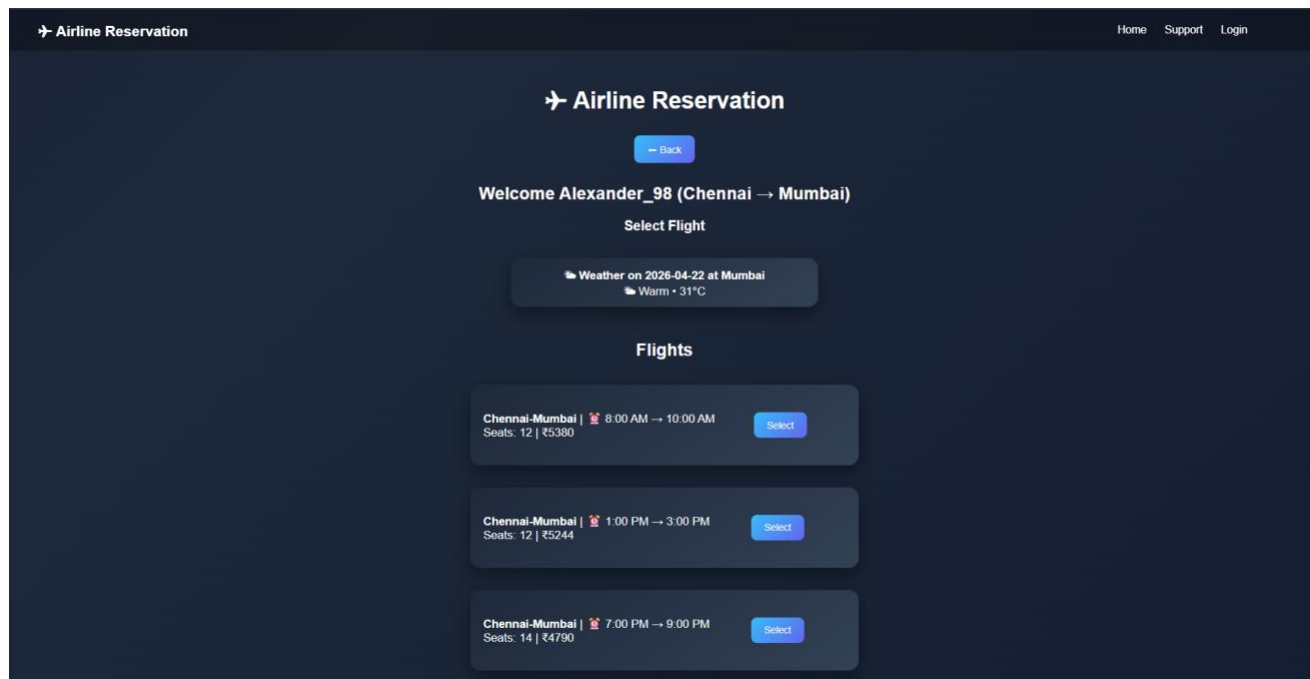


Figure 7.4 Flight Selection Page

7.5. Seat Selection Page

The figure 7.5 shows the seat selection page of the Airline Reservation System. After selecting a flight, the system displays a graphical seat layout representing the seating arrangement inside the aircraft. The seats are organized in rows and columns, with each seat labeled using a combination of row number and column letter. Different colors are used to indicate seat status, where green represents available seats, red indicates already booked seats, and yellow shows the seats selected by the user.

Users can interactively click on available seats to select or deselect them. This visual representation helps users easily identify their preferred seats such as window, aisle, or middle positions. The system also ensures that already booked seats cannot be selected again.

Once the desired seats are selected, the user can proceed to the next step by clicking the “Next” button. This page enhances the user experience by providing a clear and intuitive way to choose seats before confirming the booking.

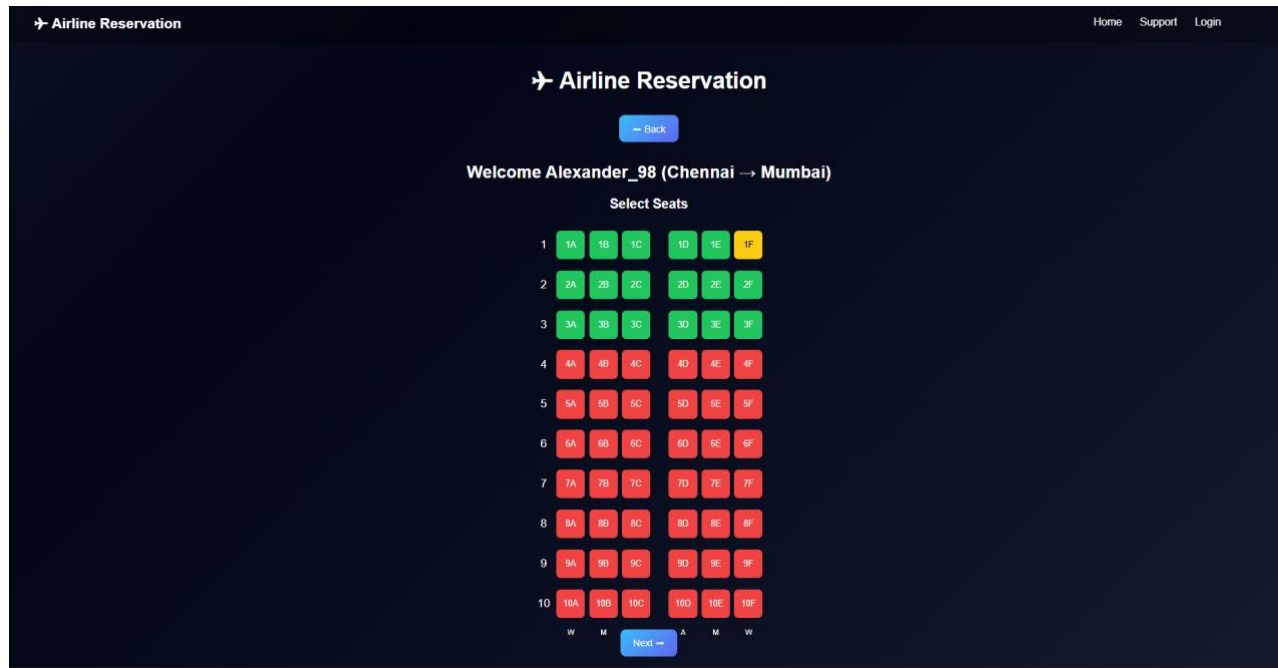


Figure 7.5 Seat Selection Page

7.6. Booking Details Page

The figure 7.6 shows the booking details page of the Airline Reservation System. After selecting the seats, the user is required to provide additional booking preferences before confirming the ticket.

This page allows the user to select the trip purpose, such as Vacation, Business, or Emergency. Each option influences the final ticket price, where certain choices may increase or decrease the cost. The user also selects the booking type, which can be either Safe or Risk. The Risk option offers a discounted price but introduces uncertainty in the booking outcome, while the Safe option follows standard booking conditions.

Once the user selects both the trip purpose and booking type, they can proceed by clicking the “Confirm Booking” button. The system then calculates the final price based on these selections and moves to the payment stage.

This page is important as it introduces dynamic pricing and personalization into the booking process, making the system more realistic and interactive.

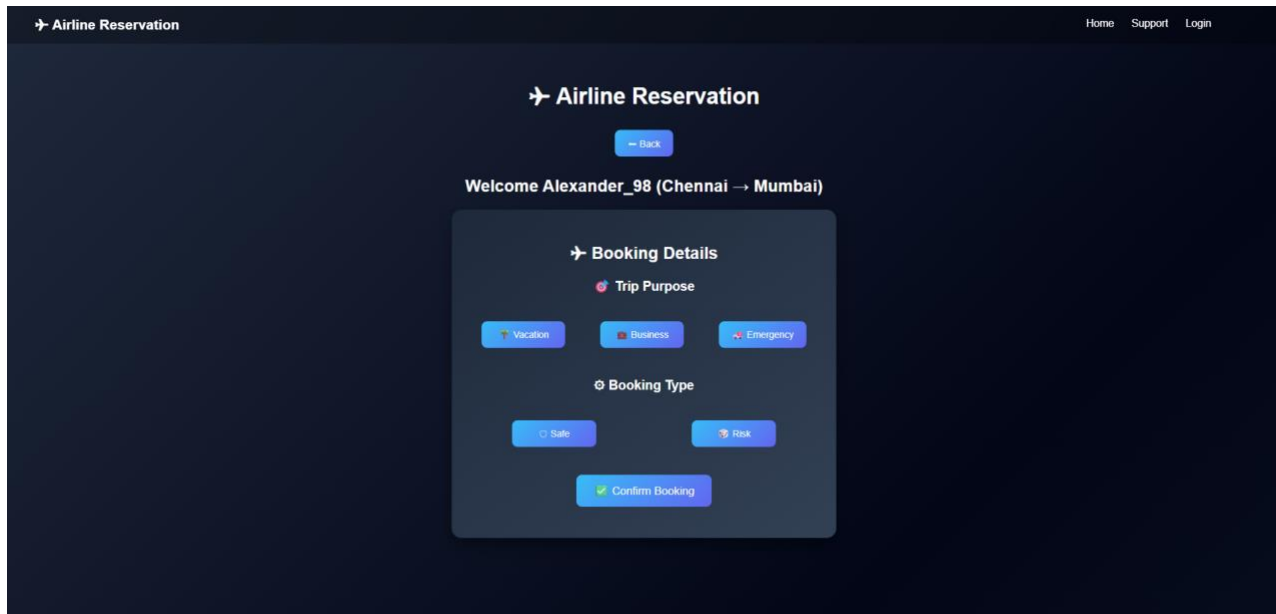


Figure 7.6 Booking Details Page

7.7 Payment Page

The figure 7.7 shows the payment page of the Airline Reservation System. After confirming the booking details, the user is directed to this page to complete the transaction. The interface provides input fields for entering card details such as card number, cardholder name, and CVV. These inputs are validated to ensure that all required fields are filled before proceeding. Once the user clicks the “Pay Now” button, the system simulates the payment process. After successful validation and processing, the system confirms the payment and proceeds to generate the ticket. This page represents the final step before booking confirmation and ensures that the transaction flow is completed smoothly within the system.

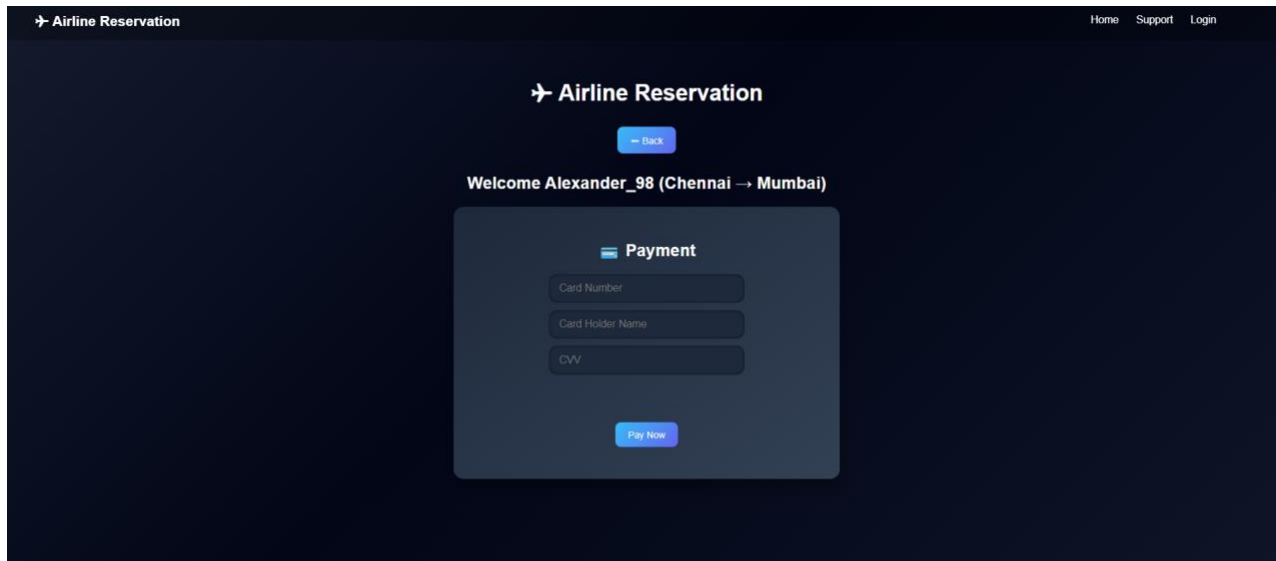


Figure 7.7 Payment Page

7.8 Ticket Generation Page

The figure 7.8 shows the ticket generation page of the Airline Reservation System, which is displayed after successful payment. This page represents the final output of the booking process and contains all the necessary details of the confirmed ticket. The ticket includes passenger information such as name, route, source, destination, travel date, and selected seat number. It also displays pricing details, including the base price and the final price after applying dynamic pricing factors like trip purpose and booking type.

On the right side, additional details such as trip purpose, booking type, and booking status are shown. The system clearly indicates whether the booking is normal, upgraded, or affected by risk conditions. The page also provides options to download the ticket or cancel the booking. An important feature of this page is the option to download the ticket. When the user clicks the “Download Ticket” button, the system generates a text file containing all booking details. This allows the user to save or print the ticket for future reference, similar to a real airline e-ticket.

The page also provides a “Cancel Ticket” option. If the user chooses to cancel the booking, the system processes a refund based on the booking type. For safe bookings, a higher refund

percentage is provided, while risk bookings offer a lower refund. After cancellation, the booking status is updated, and the refund amount is displayed to the user.

The download option allows users to save their ticket for future reference, while the cancellation option enables users to request a refund based on the system's refund policy. This page completes the booking workflow and provides confirmation to the user that the ticket has been successfully generated.

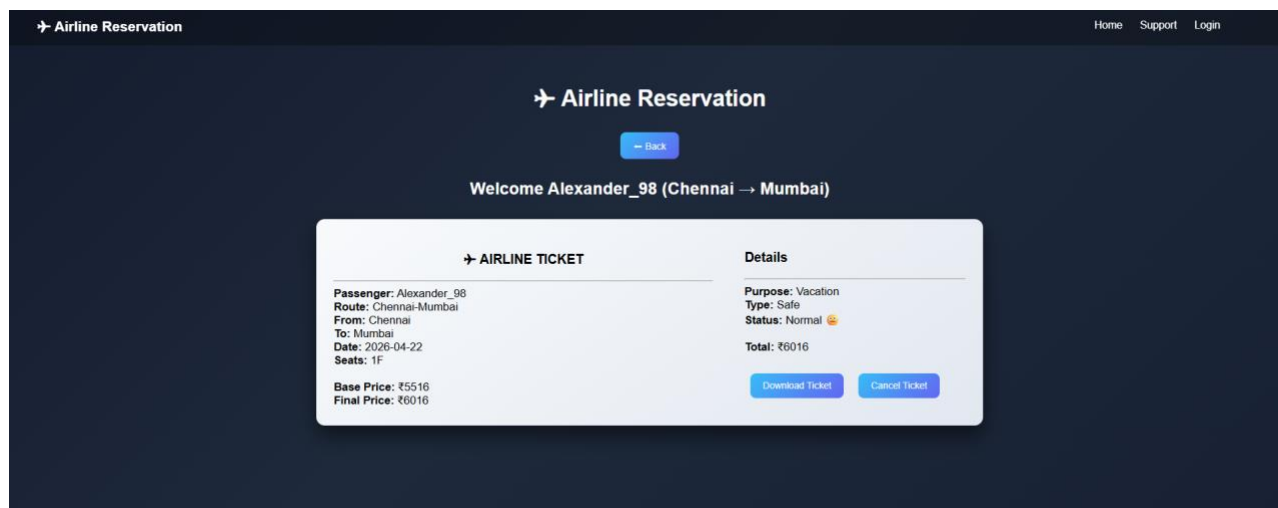


Figure 7.8 Ticket Generation Page

7.9 Ticket Cancellation and Refund Page

The figure 7.9 shows the ticket cancellation and refund page of the Airline Reservation System. This page is displayed when the user chooses to cancel a previously booked ticket.

Once the cancellation is initiated, the system updates the booking status to “Cancelled” and calculates the refund amount based on the selected booking type. The refund amount is then displayed clearly along with the status marked as “Refunded,” indicating that the cancellation process has been completed successfully.

The interface provides immediate feedback to the user, confirming that the ticket has been cancelled and the refund has been processed. This feature enhances user convenience by allowing

them to manage their bookings even after confirmation and ensures transparency in the refund process.

This page represents the final stage of the cancellation workflow and demonstrates the system's ability to handle post-booking operations effectively.

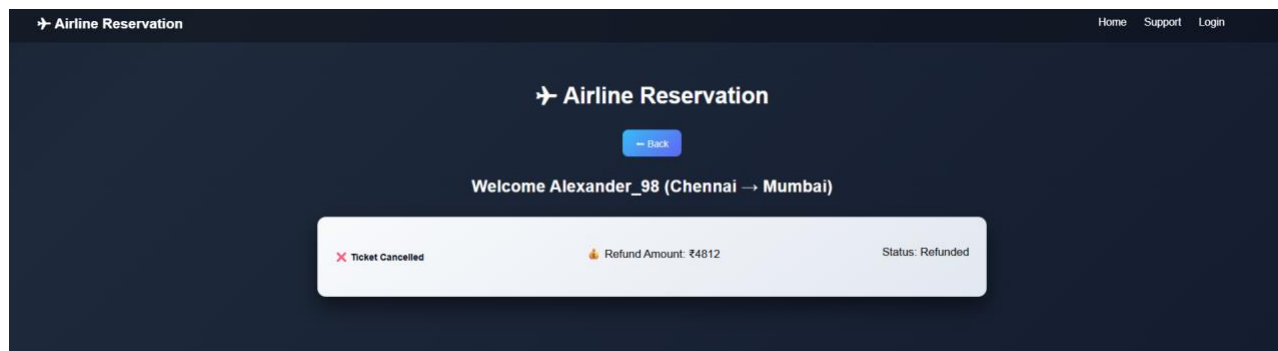


Figure 7.9 Ticket Cancellation & Refund

7.10 Support Page

The figure 7.10 shows the support page of the Airline Reservation System. This page provides users with a simple way to contact the system for any queries, issues, or assistance related to bookings. The interface displays a support email address, which is presented as a clickable link. When the user clicks on the email link, it redirects them to their default email service (such as Gmail) with the recipient address pre-filled. This allows users to quickly send their queries without manually typing the email. The page is minimal and focused, ensuring that users can easily find contact information without confusion. It enhances the overall usability of the system by providing a dedicated support channel for user assistance.

The interface is intentionally kept minimalistic and user-friendly, ensuring that users can easily locate the support contact information without unnecessary distractions. The page prominently displays the official support email address, which is implemented as a clickable hyperlink. When the user clicks on this link, it automatically redirects to their default email client (such as Gmail or

Outlook) with the recipient address pre-filled. This feature reduces user effort by eliminating the need to manually enter the email address, thereby improving usability and convenience.

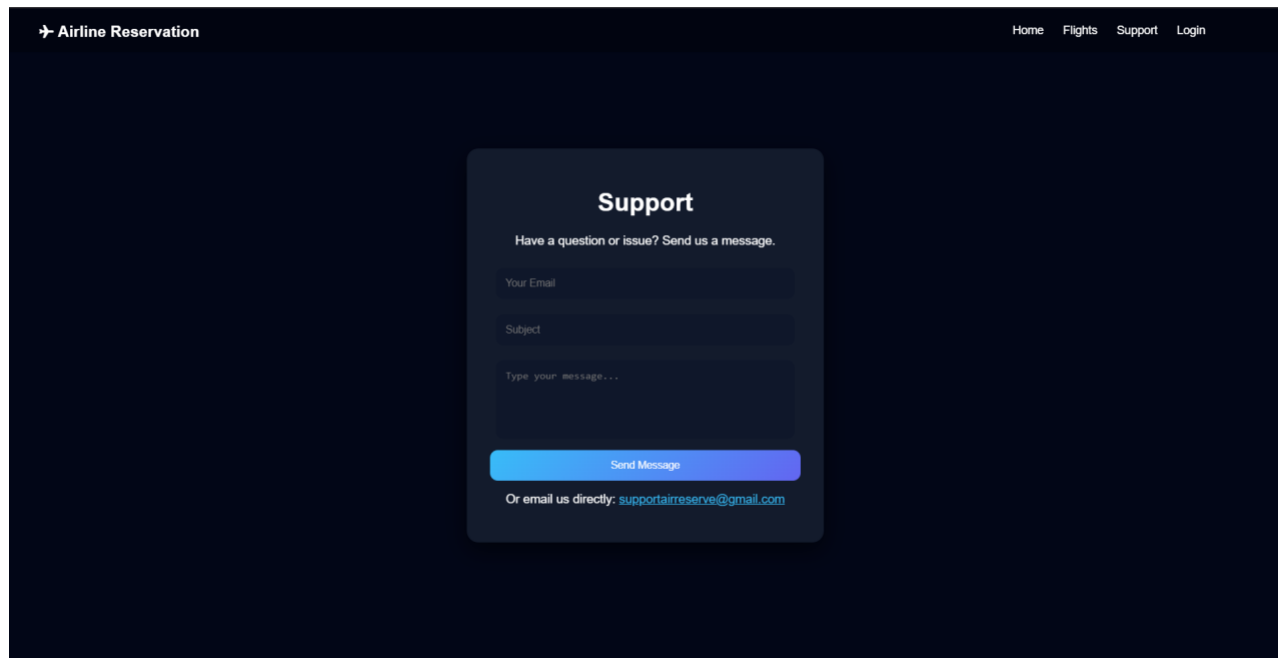


Figure 7.10 Support Page

7.11. OVERALL OUTPUT ANALYSIS

The Airline Reservation System successfully demonstrates the complete workflow of flight booking, including user authentication, flight search, seat selection, payment simulation, and ticket generation. All modules function correctly and are well integrated, providing a smooth user experience.

Key observations from testing:

- The authentication system correctly manages user login and access.
- Flight search dynamically generates options based on selected cities.
- Seat selection prevents duplicate bookings and maintains accuracy.
- The booking and payment module works efficiently, leading to successful ticket generation.

- The cancellation and refund feature updates seat availability and calculates refunds correctly.
- The support page provides a simple and effective way for users to contact support.

Overall, the system is reliable, user-friendly, and demonstrates an effective implementation of a web-based airline reservation system.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

The project titled "Airline Reservation System with Dynamic Pricing and Seat Selection" has been successfully designed and implemented. The system provides a complete flight booking solution with an interactive user interface and modular design. The application allows users to register, log in, search flights, select seats, apply booking preferences, simulate payments, and generate tickets. The system also incorporates dynamic pricing based on travel purpose and booking type (Safe or Risk), making it more realistic and user-oriented. The seat selection feature provides a visual representation of available and booked seats, improving user experience. The system also includes cancellation and refund functionality, ensuring flexibility for users. The objectives defined at the beginning of the project have been successfully achieved. A fully functional booking system was developed, user authentication was implemented, dynamic pricing logic was integrated, and complete booking flow from search to ticket generation was realized.

8.2 FUTURE WORK

1. Payment Gateway Integration

Real payment gateways such as Razorpay or Stripe can be integrated to replace the current simulated payment system.

2. Real-Time Notifications

Notifications can be added to inform users about booking confirmation, cancellations, or flight updates instantly.

3. Database Integration

The system can be upgraded from file storage and localStorage to a full database like MySQL for better scalability and data management.

4. Mobile Application

A mobile app version using technologies like Flutter or React Native can be developed for better accessibility.

5. Flight Management System

Admin functionality can be enhanced to manage real-time flight schedules, delays, and cancellations.

6. User Profile and History

Users can be provided with booking history, travel records, and personalized recommendations.

7. Advanced Seat Selection

Seat selection can be improved with categories like window, aisle, and premium seating.

8. AI-Based Recommendations

The system can suggest flights based on user preferences, previous bookings, and travel patterns using machine learning techniques.

9. Multi-User Support and Security

Improved authentication mechanisms and role-based access control can be implemented for better security.

8.3 Limitations

1. No Real Payment System

The system currently simulates payment and does not support real online transactions.

2. Limited Data Storage

User data is stored using localStorage and text files, which is not suitable for large-scale deployment.

3. No Real-Time Flight Data

Flights are generated dynamically and do not reflect real airline schedules.

4. Web-Based Only System

The application is accessible only through a web browser and lacks a mobile application.

5. Limited Scalability

The system is designed for small-scale usage and may not handle large numbers of users efficiently.

APPENDIX 1: SAMPLE CODE

BACKEND – AirlineReservation.java

```
import java.util.*;
import java.io.*;
class Flight {
    int id, seats, price;
    String route;
    Flight(int id, String route, int seats, int price) {
        this.id = id;
        this.route = route;
        this.seats = seats;
        this.price = price;
    }
}
class Passenger {
    String name, preferredRoute;
    int score, trips, cancellations;
    Passenger(String n, int s, int t, int c, String r) {
        name = n; score = s; trips = t;
        cancellations = c; preferredRoute = r;
    }
}
public class AirlineReservation {
    static List<Flight> flights = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        flights.add(new Flight(1,"Chennai-Delhi",3,5000));
        flights.add(new Flight(2,"Chennai-Mumbai",2,4500));
        System.out.print("Enter name: ");
```



```

Passenger user = loadUser(sc.nextLine());
while(true){
    System.out.println("\n1.View 2.Book 3.Cancel 4.Exit");
    int ch = sc.nextInt();
    if(ch==1) viewFlights();
    else if(ch==2) bookTicket(user);
    else if(ch==3) cancelTicket(user);
    else { saveUser(user); break; }
}
}

static Passenger loadUser(String name){
    try(BufferedReader br=new BufferedReader(new FileReader("users.txt"))){
        String line;
        while((line=br.readLine())!=null){
            String[] d=line.split(",");
            if(d[0].equalsIgnoreCase(name))
                return new Passenger(d[0],Integer.parseInt(d[1]),
                    Integer.parseInt(d[2]),Integer.parseInt(d[3]),d[4]);
        }
    } catch(Exception e){}
    return new Passenger(name,0,0,0,"None");
}

static void saveUser(Passenger u){
    try(BufferedWriter bw=new BufferedWriter(new FileWriter("users.txt",true))){
        bw.write(u.name+","+u.score+","+u.trips+","+u.cancellations+","+u.preferredRoute);
        bw.newLine();
    } catch(Exception e){}
}

static void viewFlights(){
    for(Flight f:flights)
        System.out.println(f.id+" "+f.route+" | Seats:"+f.seats+" | ₹"+f.price);
}

```

```

    }
    static void bookTicket(Passenger u){
        viewFlights();
        System.out.print("Select ID: ");
        Flight f = flights.stream()
            .filter(x->x.id==sc.nextInt()).findFirst().orElse(null);
        if(f==null || f.seats==0){
            System.out.println("Invalid/Full flight"); return;
        }
        int price = f.price;
        System.out.println("Purpose:1.Vac 2.Bus 3.Emerg");
        int p = sc.nextInt();
        if(p==1) price+=500;
        else if(p==3) price-=1000;
        System.out.println("Type:1.Safe 2.Risk");
        if(sc.nextInt()==2)
            System.out.println(new Random().nextBoolean() ?
                "👤 Upgrade" : "🚫 Last seat");
        f.seats--; u.trips++;
        u.preferredRoute = f.route;
        System.out.println("✅ Booked | "+f.route+" | ₹"+price);
    }
    static void cancelTicket(Passenger u){
        u.cancellations++;
        System.out.println("❌ Cancelled");
    }
}

```

home.html

```
<!DOCTYPE html>
```

```
<html>
```

```

<head>
<title>Airline</title>
<style>
body {
    margin: 0;
    padding-top: 80px;
    font-family: Arial;
    color: white;
    background: linear-gradient(-45deg, #020617, #0f172a, #1e293b);
    height: 100vh;
}
/* TOP NAVBAR */
.navbar {
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    padding: 15px 40px;
    display: flex;
    justify-content: space-between; /* keeps logo left, menu right */
    align-items: center;
    box-sizing: border-box;
    background: rgba(0,0,0,0.3);
    backdrop-filter: blur(10px);
}
/* LOGO */
.logo {
    font-size: 20px;
    font-weight: bold;
}
/* BUTTON GROUP */

```

```

.nav-buttons {
  display: flex;
  gap: 160px;
  margin-left: 380px; /* 🔥 increase this */
}

.nav-buttons a {
  text-decoration: none;
  color: white;
  font-size: 15px;
  font-weight: 500;
  transition: 0.3s ease;
}

/* 🔥 Hover effect */
.nav-buttons a:hover {
  color: #38bdf8;
}

.nav-buttons button:hover {
  transform: translateY(-2px);
  box-shadow: 0 5px 15px rgba(56,189,248,0.4);
}

/* HERO SECTION */
.hero {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
}

.hero-content {
  text-align: center;
}

.hero h1 {

```

```

    font-size: 48px;
    color: aliceblue
}
.hero p {
    color: #94a3b8;
}
.hero button {
    margin-top: 20px;
    padding: 12px 25px;
    border-radius: 10px;
    background: linear-gradient(135deg, #38bdf8, #6366f1);
    color: white;
    cursor: pointer;
}
.bottom-planes {
    position: fixed;
    bottom: 0;
    width: 100%;
    height: 150px;
    pointer-events: none;
    overflow: hidden;
}
.bottom-planes img {
    position: absolute;
    bottom: 10px;
    opacity: 0.2;
    animation: flyBottom linear infinite;
}
@keyframes flyBottom {
    0% {
        left: -50px;
    }

```

```

        transform: translateY(0) rotate(5deg);
    }
    50% {
        transform: translateY(-10px) rotate(2deg);
    }
    100% {
        left: 110%;
        transform: translateY(0) rotate(5deg);
    }
}
</style>
</head>
<body>
<!-- NAVBAR -->
<div class="navbar">
    <div class="logo">✈ Airline Reservation</div>
    <div class="nav-buttons">
        <a href="home.html">Home</a>
        <a href="support.html">Support</a>
        <a href="login.html">Login</a>
    </div>
</div>
<!-- 🦋 HERO -->
<div class="hero">
    <div class="hero-content">
        <h1>Fly Smart. Fly Easy.</h1>
        <p>Book flights, choose seats, and travel smarter.</p>
        <button onclick="goLogin()">Get Started</button>
    </div>
</div>
<div class="bottom-planes"></div>

```

```

<script>
function goLogin(){
    window.location.href = "login.html";
}
</script>
<script>
function goHome(){
    window.location.href = "home.html";
}
function goLogin(){
    window.location.href = "login.html";
}
function goApp(){
    window.location.href = "index.html";
}
function goSupport(){
    window.location.href = "support.html";
}
</script>
</body>
</html>

```

index.html

```

import <!DOCTYPE html>
<html>
<head>
<title>Airline System</title>
<style>
body{
    padding-top:80px;
    font-family:Arial;

```

```

    color:white;
    text-align:center;
    background:linear-gradient(-45deg,#0f172a,#1e293b,#020617);
}
.card{
    background:#1e293b;
    padding:15px;
    margin:15px auto;
    border-radius:10px;
    width:450px;
}
input,select{
    padding:10px;
    margin:8px;
    border-radius:8px;
    border:none;
    background:#0f172a;
    color:white;
}
button{
    padding:10px 15px;
    border:none;
    border-radius:8px;
    background:#38bdf8;
    color:white;
    cursor:pointer;
}
</style>
</head>
<body>
<h1>✈️ Airline Reservation</h1>

```



```

<div id="homePage">
<div class="card">
  <input id="name" placeholder="Name">
  <select id="from">
    <option>Chennai</option>
    <option>Delhi</option>
    <option>Mumbai</option>
  </select>
  <select id="to">
    <option>Chennai</option>
    <option>Delhi</option>
    <option>Mumbai</option>
  </select>
  <input type="date" id="date">
  <button onclick="start()">Start</button>
</div>
</div>
<div id="main" style="display:none;">
<h2 id="welcome"></h2>
<div id="flights"></div>
<div id="seatMap"></div>
<div id="ticket"></div>
</div>
<script>
let flights=[];
let selectedFlight=null,selectedSeats=[];
let fromCity,toCity,travelDate,finalPrice=0;
function start(){
  fromCity=from.value;
  toCity=to.value;
  travelDate=date.value;

```

```

    if(fromCity===toCity) return alert("Invalid cities");
    document.getElementById("homePage").style.display="none";
    document.getElementById("main").style.display="block";
    showFlights();
}
function showFlights(){
    let timings=["8AM","1PM","7PM"];
    flights=timings.map((t,i)=>({
        id:i+1,
        route:fromCity+"-"+toCity,
        time:t,
        price:4000+Math.random()*2000|0
    }));
    flightsDiv.innerHTML=flights.map(f=>`
        <div class="card">
            ${f.route} | ${f.time} | ₹${f.price}
            <button onclick="selectFlight(${f.id})">Select</button>
        </div>
    `).join("");
}
function selectFlight(id){
    selectedFlight=flights.find(f=>f.id===id);
    seatMap.innerHTML="Select Seats...";
}
function confirmBooking(){
    finalPrice=selectedFlight.price;
    alert("Booking Confirmed ₹"+finalPrice);
}
</script>
</body>
</html>

```

login.html

```
<!DOCTYPE html>
<html>
<head>
<title>Login - Airline</title>
<style>
body{
  margin:0;
  font-family:Arial;
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
  background:#0f172a;
  color:white;
}
.login-box{
  background:#1e293b;
  padding:30px;
  border-radius:12px;
  width:320px;
  text-align:center;
}
input{
  width:90%;
  padding:10px;
  margin:10px 0;
  border-radius:8px;
  border:none;
  background:#020617;
  color:white;
```

```

}
button{
    width:100%;
    padding:10px;
    border:none;
    border-radius:8px;
    background:#38bdf8;
    color:white;
    cursor:pointer;
}
.switch{
    margin-top:10px;
    font-size:14px;
    color:#94a3b8;
    cursor:pointer;
}
</style>
</head>
<body>
<div class="login-box">
    <h2 id="title">Login</h2>
    <input id="username" placeholder="Username">
    <input type="password" id="password" placeholder="Password">
    <button onclick="handleAuth()">Login</button>
    <div class="switch" onclick="toggleMode()">
        Don't have an account? <span>Register</span>
    </div>
</div>
<script>
let isLogin=true;
function toggleMode(){

```

```

isLogin=!isLogin;
title.innerText=isLogin?"Login":"Register";
document.querySelector("button").innerText=isLogin?"Login":"Register";
document.querySelector(".switch").innerHTML=isLogin?
`Don't have an account? <span>Register</span>`:
`Already have an account? <span>Login</span>`;
}
function handleAuth(){
  let user=username.value, pass=password.value;
  if(!user||!pass) return alert("Fill all fields");
  let users=JSON.parse(localStorage.getItem("users"))||{};
  if(isLogin){
    if(users[user]===pass){
      localStorage.setItem("currentUser",user);
      window.location="index.html";
    } else alert("Invalid credentials");
  } else {
    if(users[user]) return alert("User exists");
    users[user]=pass;
    localStorage.setItem("users",JSON.stringify(users));
    alert("Registered!");
    toggleMode();
  }
}
</script>
</body>
</html>

```

support.html

```

<!DOCTYPE html>
<html>
<head>

```

```
<title>Support</title>
<style>
body{
  margin:0;
  padding-top:80px;
  font-family:Arial;
  color:white;
  background:#020617;
}
.navbar{
  position:fixed;
  width:100%;
  padding:15px 40px;
  display:flex;
  justify-content:space-between;
  align-items:center;
  background:rgba(0,0,0,0.3);
  backdrop-filter:blur(10px);
}
.logo{
  font-size:20px;
  font-weight:bold;
}
.nav-buttons{
  display:flex;
  gap:25px;
}
.nav-buttons a{
  text-decoration:none;
  color:white;
  font-size:15px;
```

```

}
.nav-buttons a:hover{
    color:#38bdf8;
}
.content{
    margin-top:100px;
    text-align:center;
}
</style>
</head>
<body>
<div class="navbar">
    <div class="logo">✈ Airline Reservation</div>
    <div class="nav-buttons">
        <a href="home.html">Home</a>
        <a href="index.html">Flights</a>
        <a href="support.html">Support</a>
        <a href="login.html">Login</a>
    </div>
</div>
<div class="content">
    <h1>Support Page</h1>
    <p>Email:
        <a href="https://mail.google.com/mail/?view=cm&fs=1&to=support@airreserve.com"
            target="_blank" style="color:#38bdf8;">
            support@airreserve.com
        </a>
    </p>
</div>
</body>
</html>

```

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